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Keyless Entry Specification (Bluetooth)

Project code	N600/J107
Document Details	EOL – 2W/3W Keyless Entry Specification
Component	Cluster, VCU and EMS
Release Date	28th Aug 2025
Author	Upendra Kumar and Vinod Pendela
Reviewer	NK Naveen and Nabakrushna
Approver	Parag Ashok Inamdar

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1. Purpose

This System Requirements specification is applicable for keyless system Diagnostics.

2. Product Requirements

Vehicle Type:

Generic 2 and 3-Wheeler (2W/3W) supporting Keyless entry.

Customer profile:

Supplier shall meet and understand the customer profile from the TVS-development team

3. Product Brief

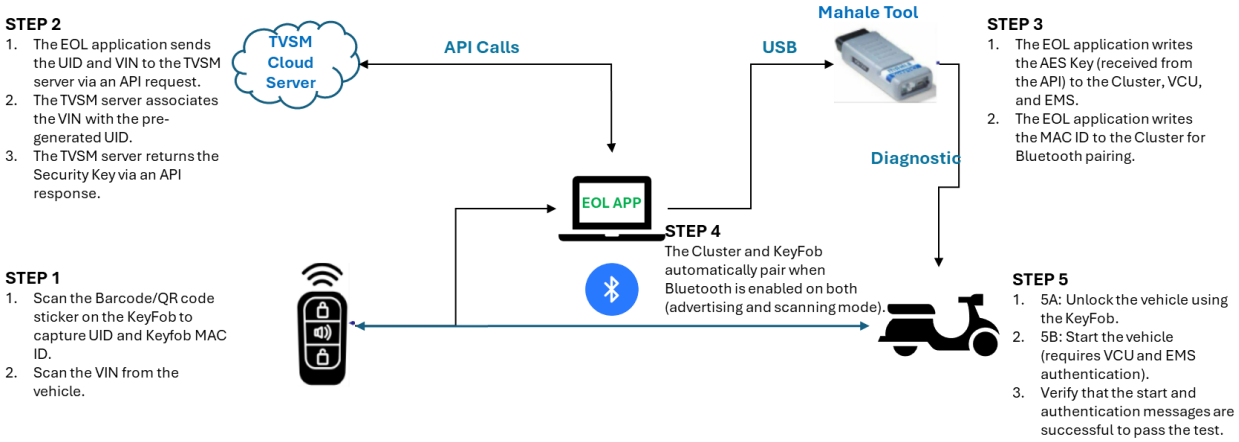
1. Suppliers shall meet and understand the product brief from the TVS-M development team

2. Diagnostic requirements for Flashing AES Keys and Keyfob MAC ID in the Vehicle

   - A Keyfob is mandatory for flashing the MAC ID into the vehicle.
   - Each Keyfob has a barcode on its key tag containing a unique device ID and MAC ID, which must be scanned at the End of Line (EOL).
   - The VIN for each vehicle must also be scanned at the EOL.
   - The EOL application software, enabled with RSA encryption, shall be used to write the AES key into the vehicle. This AES key enables authentication between the Keyfob and the vehicle when the customer attempts to access it.
   - The EOL application shall use the unique device ID and MAC ID from Keyfob, along with the VIN, as inputs to the TVSM cloud server. The application shall then fetch the AES key from the cloud server, which are mapped to that specific Keyfob in the TVSM cloud.
   - Upon receiving the AES key from the TVSM cloud server, the diagnostic tool shall unlock the Cluster, VCU, and EMS It shall then write the AES key into the Cluster, VCU, and EMS. It will also write the MAC ID into the Cluster.

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4. Workflow Diagram



5. DID Information

DID Type	DID Value	Components	UDS Spec Version
MAC Write	F1 38	Cluster	11.2.3
MAC Read	F1 38	Cluster	11.2.3
AES Key	F1 39	Cluster, VCU, EMS	11.2.3

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6. UDS Workflow

6.1. Tester - Cluster

	Action	Diagnostic Service	Positive Response
Step 1	Default session	10 01	50 01 XX XX XX XX
Step 2	Switch to Extended Session	10 03	50 03 XX XX XX XX
Step 3	Request Seed	27 01	67 01 XX XX XX XX XX XX XX XX
Step 4	Send Key	27 02 XX XX XX XX XX XX XX XX	67 02
Step 5	Erase all previously paired keys	31 01 4F 17	71 01 4F 17 00
Step 6	Write AES Key	2E F1 39 XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX	6E F1 39
Step 7	Write MAC ID(s)	2E F1 38 NN XX XX XX XX XX XX	6E F1 38
Step 8	Read MAC ID	22 F1 38	62 F1 38 NN XX XX XX XX XX XX

NN – Number of key fobs to be paired

Note: The example Diagnostic messages mentioned in the steps 7 and 8 are given with NN = 1

6.2. Tester - VCU

	Action	Diagnostic Service	Positive Response
Step 1	Default session	10 01	50 01 XX XX XX XX
Step 2	Switch to Extended Session	10 03	50 03 XX XX XX XX
Step 3	Request Seed	27 01	67 01 XX XX XX XX XX XX XX XX
Step 4	Send Key	27 02 XX XX XX XX XX XX XX XX	67 02
Step 5	Write AES Key	2E F1 39 XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX	6E F1 39

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6.3. Tester - EMS

	Action	Diagnostic Service	Positive Response
Step 1	Default session	10 01	50 01 XX XX XX XX
Step 2	Switch to Extended Session	10 03	50 03 XX XX XX XX
Step 3	Request Seed	27 01	67 01 XX XX XX XX XX XX XX XX
Step 4	Send Key	27 02 XX XX XX XX XX XX XX XX	67 02
Step 5	Write AES Key	2E F1 39 XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX	6E F1 39

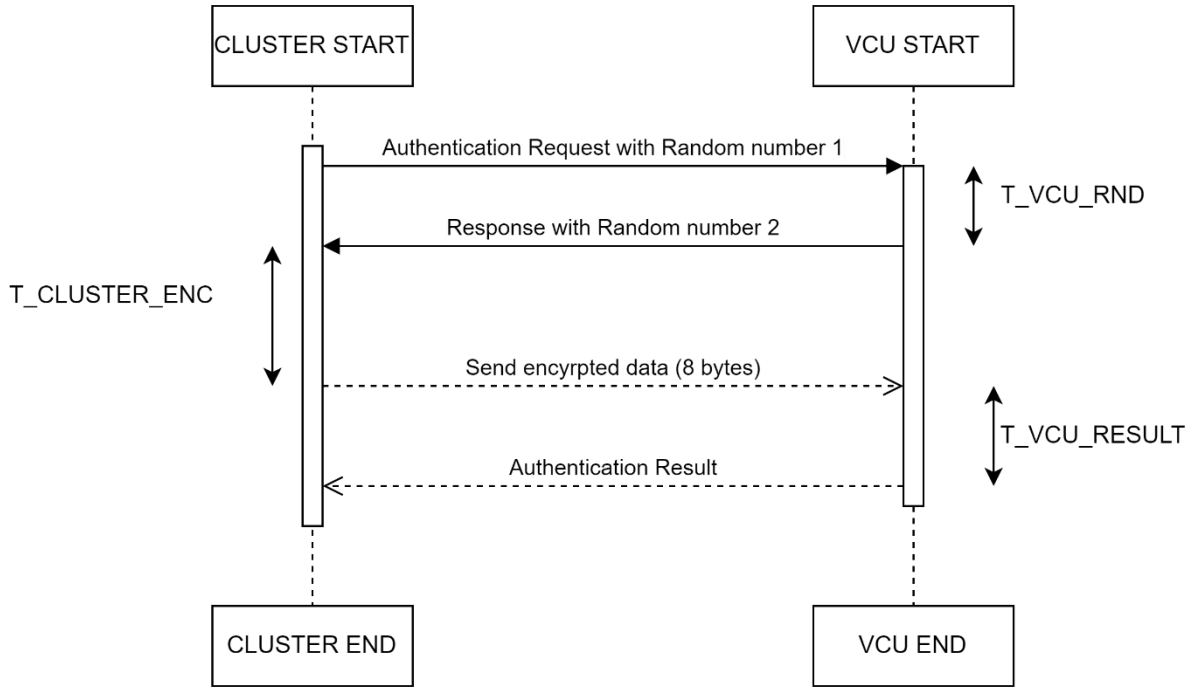
7. Authentication Flow

The Authentication process (via CAN) among VCU, Cluster and EMS would start after Ignition ON.

7.1. VCU-CLUSTER Authentication

1. Cluster initiates the authentication by transmitting a 4-byte random number.
2. VCU responds with 4-byte random number within pre-defined time (T_VCU_RND).
3. Both the ECUs calculate the 16 Byte encrypted data using the random numbers generated by Cluster and VCU and 8 byte fixed data.
4. Cluster sends 8 bytes (from the calculated encrypted data) to VCU.
5. VCU validates the data received and responds with Authentication status.
6. If there is a time out in message reception, there shall be 3 retries for the authentication process. If still unsuccessful, authentication process shall be terminated.

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7.2. VCU-EMS Authentication

1. VCU shall send an authentication request to EMS ECU.
2. EMS shall send a 6-byte random number to VCU within pre-defined time (T_EMS_RND) and if VCU does not receive the random number then it will re-initiate authentication request.

3. EMS will calculate the 16 Byte encrypted data using the same random number sent to VCU and 10 Byte fixed data.

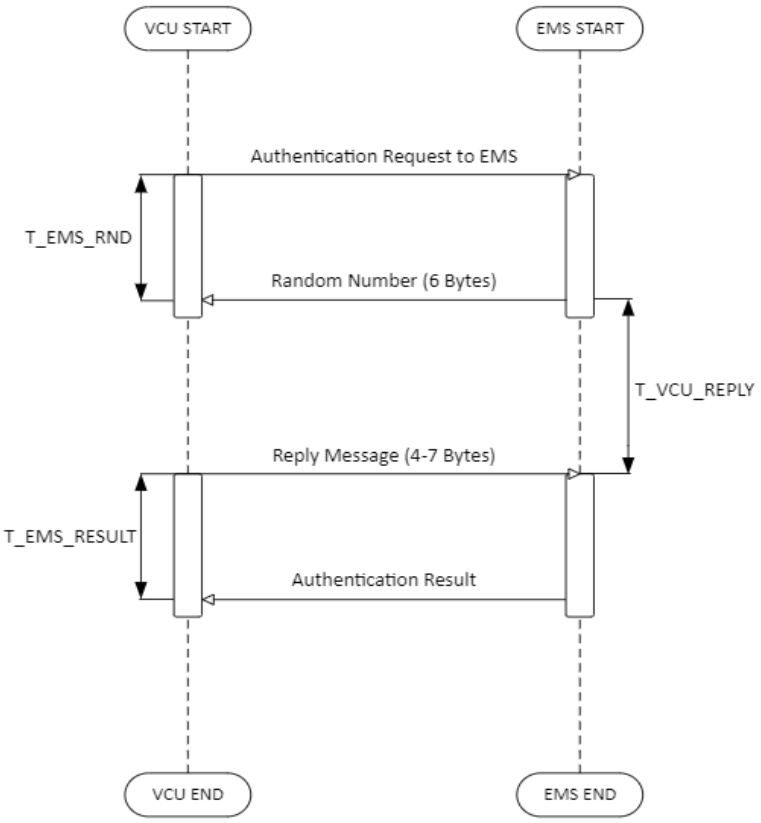
4. Using the 6-byte random number and 10 Byte fixed data VCU will calculate reply data according to AES 128 authentication algorithm to generate 16 Byte data.

5. Then VCU will send Byte number 4-7 of encrypted output data to EMS through reply message.

6. EMS will then compare the 4-7 byte of EMS encrypted output data and reply data from VCU and send authentication result to VCU.

7. If there is a time out in message reception, there shall be 3 retries for the authentication process. If still unsuccessful, authentication process shall be terminated.

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8. References

1. TVSM_Diagnostic_Specification.pdf
2. Secure Key Fob Management (API User Manual for AES key generation)